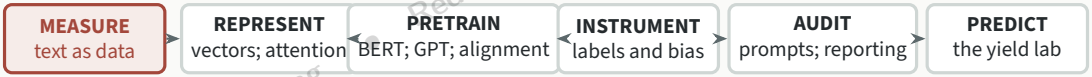
Session 5.1

Text as Economic Data: From Documents to Counts

Clean, tokenize, count: how a document becomes a vector a regression can use

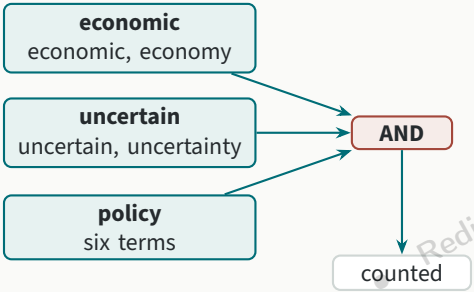
Based on Chapter 13 of Miao & Feng, *AI for Economists*



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Reading Policy Uncertainty Out of Newspapers

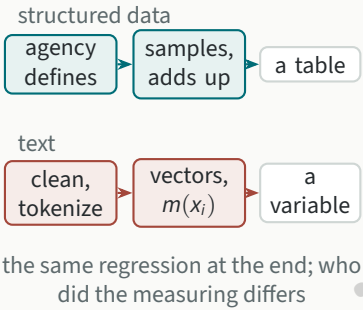


an article counts only if it hits all three groups; the same recipe gives China's index from 1949 (Davis, Liu & Sheng 2019)

- Baker, Bloom & Davis (2016): ten US newspapers, monthly since 1985; an article counts if it hits all three groups.
- $U_t = \frac{1}{10} \sum_p \frac{c_{pt}/N_{pt}}{\sigma_p}$, with c_{pt} hits among N_{pt} articles of paper p; the index is $100\, U_t / \bar{U}$.
- Checked against 12,009 hand-coded articles: machine and human indices correlate 0.86 by quarter (1985–2012), 0.93 by year (1900–2010).
- **No agency publishes this number: it was built from words and checked against readers.**

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With Text, the Measurement Step Is Yours

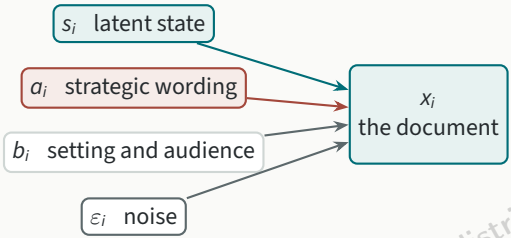


text is the	example	study
measure	policy uncertainty from newspapers	Baker, Bloom & Davis (2016)
outcome	FOMC speech after transcripts went public	Hansen, McMahon & Prat (2018)
treatment	central-bank talk and inflation expectations	Blinder et al. (2024)

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- Prices arrive measured; statements, filings and job ads arrive as language. For a price index an agency defines, samples and adds up; for text nobody has.
- Text can be the regressor, the outcome or the treatment; this session builds the first.
- **With text, the researcher does the measuring.**

What a Text Measure Measures



two firms, the same risk s_i: one discloses, one deflects, so x_i differs through a_i

- The document is not the state: $x_i = g(s_i, a_i, b_i) + \varepsilon_i$. A measure $m(x_i)$ then enters $y_i = \beta\, m(x_i) + \boldsymbol{\gamma}^\top \mathbf{c}_i + u_i$.
- If the wording a_i shares a driver with u_i, $\hat{\beta}$ is biased, and a bigger corpus does not repair it.
- The measurement error $m(x_i) - s_i$ moves with s_i itself (a firm hiding a risk writes less about it): not classical, so no attenuation rule, and the sign of the bias is unknown.
- **Ask why $m(x_i)$ is exogenous before asking how well it is measured.**

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The Pipeline: Four Steps From a String to a Vector



	after step 1: clean	after step 2: tokenize	after step 3: count
EN	The central bank raised interest rates amid rising inflation.	The · central · bank · raised · interest · rates · amid · rising · inflation · .	one row of V counts, mostly 0
ZH	央行上调利率以遏制通胀。	央行 · 上调 · 利率 · 以 · 遏制 · 通胀 · 。	one row of V counts, mostly 0

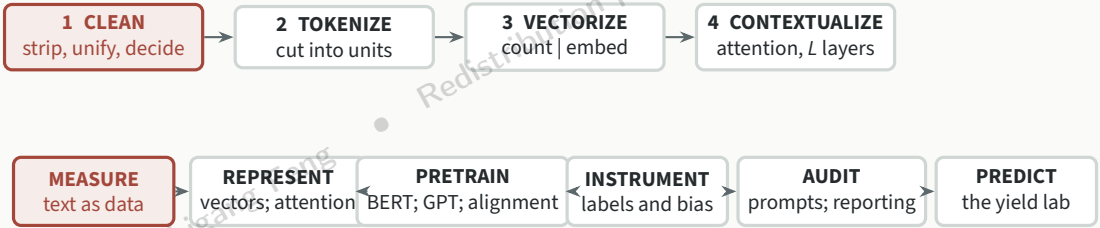
- Cleaning and tokenizing decide which strings count as one feature; vectorizing turns tokens into numbers. This deck counts; Deck 5.2 embeds, then adds context.
- **Different choices at any step give different regressors from the same documents.**

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Session 5.1

Step 1 · Clean

What to remove, what to unify, and what must stay



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Cleaning, Worked on One Sentence

English, scraped from a web page

`<p>The central bank raised …</p>`

↓ strip mark-up and entities

The central bank raised …

Chinese, traditional characters

`<p>`央行上調利率以遏制通脹。`</p>`

↓ 繁 → 简: 調 → 调, 脹 → 胀

央行上调利率以遏制通胀。

ＣＰＩ同比上漲２．３％

↓ 全角 → 半角, 漲 → 涨

CPI 同比上涨 2.3%

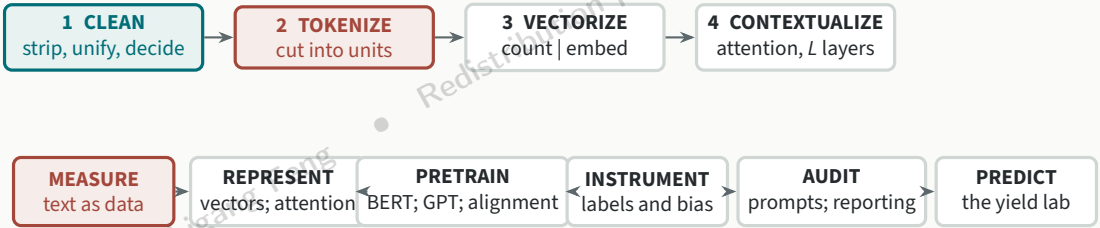
- Strip what is not text: mark-up, headers and footers, table debris, broken encodings. Unify what should be one string, or 通脹 and 通胀 are two words and ＣＰＩ never matches CPI.
- Decide by task: lower-casing merges the Fed with the verb *fed*; in finance %, ? and quotation marks carry information.
- **Write the rules as a script, and archive it with the data.**

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Session 5.1

Step 2 · Tokenize

Cut the string into the units you will count



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Tokens and the Vocabulary

English: spaces do most of the cutting

GDP	rose	by	2	%

Chinese: the boundaries must be inferred

央行	决定	下调	存款准备金率

央行决定下调存款准备金率: the central bank decides to cut the reserve requirement ratio

- A **corpus** is a collection of texts: a year of FOMC minutes, every annual report in an industry.
- A **token** is the smallest unit you count; **tokenization** cuts a string into tokens; the **vocabulary** is the set of distinct tokens.
- **Change the cut and you change every count that follows.**

EN	The	central	bank	raised	interest	rates	amid	rising	inflation	.
ZH	央行	上调	利率	以	遏制	通胀	。			

the running pair, cut into words

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Chinese Segmentation Is a Labelling Problem

研究生命科学

研究 | 生命科学 — study | life science

研究生 | 命科学 — graduate student | ?

行业政策支持

行业政策 | 支持 — head word: support

行业 | 政策支持 — head word: policy support

one label per character: B begins a word, E ends it

央	行	上	调	利	率
B	E	B	E	B	E

- Chinese characters run together, so segmentation is **sequence labelling**: for each character, decide whether a word ends after it.
- 研究生命科学 reads as *study life science*, or as nonsense if the cut falls after 生.
- 行业政策支持 splits two sensible ways with different head words: support for an industry policy, or policy support for an industry.
- **A wrong boundary changes what the sentence is about.**

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Domain Compounds Need a User Dictionary

货币政策委员会, the monetary policy committee

general dictionary

货币 | 政策委员会 head word lost

with a user dictionary

货币政策委员会 kept whole

entries to add first:

社会融资规模 | 宏观审慎监管框架

- Segmenters such as Jieba, THULAC and LTP combine a dictionary with a statistical or neural labeller.
- Domain compounds are missing from general dictionaries, and a wrong cut changes the head word.
- The fix is a **user dictionary** listing terms such as 社会融资规模 (aggregate financing), so they stay whole.
- **Add the domain terms before you count, and publish the list.**

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Stop Words: Drop the Glue, Keep the Signal

usually dropped

the is of 的 了 是

keep: they carry the signal

not up down more less

因此 therefore 然而 however

- **Stop words**: frequent function words with little topical content (the, is, of; 的, 了, 是).
- General lists also delete up, down, more and less, the direction of a macro signal, and *not*, which flips a sentence; 因此 and 然而 carry the logic.
- Domain lists differ too: *liability* is negative in a general sentiment list and a neutral line item in accounts (Loughran & McDonald 2011).
- **A general stop list can delete the signal itself.**

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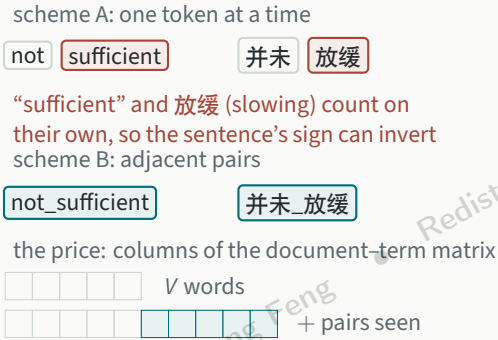
Stems, Lemmas, and a Language Without Inflection



- **Stemming** chops suffixes by rule, so investing, invests and invested all become *invest*; it also turns studies into *studi*, a non-word.
- **Lemmatization** returns the dictionary form using the part of speech: better → good, were → be.
- 上调 (raise) never changes form: tense sits in separate words, 已上调 (has raised) and 将上调 (will raise).
- **English must choose stem or lemma; Chinese has no need for either.**

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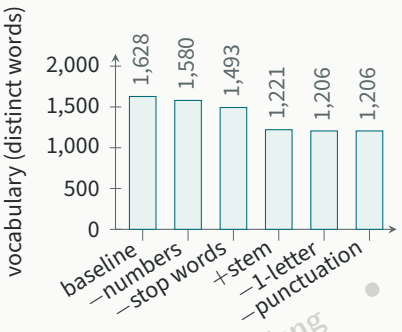
n-Grams: One Boundary Can Flip the Sign



- An **n-gram** treats n adjacent tokens as one feature: “not sufficient” stays hawkish; split, it reads as reassuring.
- Which pairs count is a rule we set: a minimum count, a collocation score, or a phrase list (Hassan et al. 2019 build one from political texts).
- The price: V words allow V^2 pairs, and every pair seen adds a column.
- **A pair is a feature when a rule says so; report the rule.**

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Six Pipelines, Five Vocabularies



step added	what it does	example
baseline	lower-case, split	Fed → fed
−numbers	drop numerals	“5-1/4” gone
−stop words	drop stop words	116: the, of, *but*
+stem	strip suffixes	inflation → infl
−1-letter	drop 1-letter tokens	stray letters
−punctuation	nothing left to drop	none

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- Each bar: the distinct words left in the 149 statements after every step up to it, 1,628 → 1,206 (−26%); the last two coincide, the tokenizer never emits punctuation.
- Each step loses something: “moderated *but* remains elevated” loses its turn with *but*.
- **Report your pipeline; show the result survives a change** (Denny & Spirling 2018).

Why Subwords

a word split into reusable pieces

macroeconomics → macro | economics

not among the lab's 1,628 words, yet spelled by its byte-pair pieces:

di·s·infl·a·tion·ar·y disinflationary

a policy term coined after the dictionary:

专精特新 → 专 | 精 | 特 | 新

- Word-level tokenizers fail twice: an unseen word gets no representation, and the vocabulary grows with every new form.
- **Subword tokenization** keeps reusable fragments, so firm names, tickers and new policy terms such as 专精特新 are spelled from known pieces.
- **With byte-level subwords, every string has a representation.**

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Why Learn the Pieces From Counts

one word, four vocabulary sizes V. Top: every string spelled, but long, and no unit means anything alone

V

$V = 256$ bytes
d·i·s·i·n·f·l·a·t·i·o·n·a·r·y 15 tokens

letters + 200 merges (the lab's BPE)
di·s·infl·a·tion·ar·y 7

cl100k, $V \approx 100{,}000$
disin·flation·ary 3

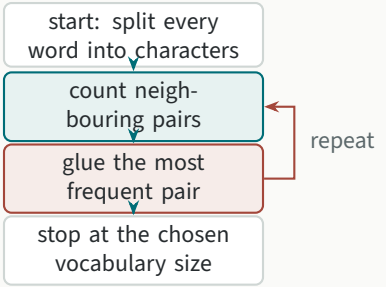
whole words, the statements' 1,628
disinflationary no slot

bottom: short, one unit one meaning, but the unseen has no slot

- Letters alone spell anything, but a statement becomes thousands of units, none meaning anything by itself.
- Whole words mean something, but an unseen word has no slot and the list never ends.
- In between, V is a choice: which strings get a slot? A language model learns one vector per slot, from that token's occurrences, so a slot pays only for a string that recurs. Counts know which.
- **Let the corpus's own counts pick the strings worth a slot.**

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Byte-Pair Encoding: Learn the Pieces From Counts



通胀 $\times 5$, 通胀率 $\times 3$, 利率 $\times 4$ (inflation, inflation rate, interest rate)

start	通	胀	通	胀	率	利	率	
merge 1	通胀		通胀	率		利	率	通+胀: $5+3=8$
merge 2	通胀		通胀	率		利率		利+率: 4; 通胀+率: 3

a new word replays the merges: 通胀目标 (inflation target) → 通胀 · 目 · 标

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- Goal: the recurring strings get a slot; any string stays spellable (Sennrich, Haddow & Birch 2016).
- What we do: start from characters; count neighbouring pairs; glue the most frequent pair (the merge that saves the most tokens); count again; repeat to the chosen size.
- **Frequency decides the cuts, not meaning: 200 merges give un·emplo·y·ment.**

Three Ways to Learn the Pieces

scheme	what it does, and what it is after	used by
byte-pair encoding	glue the most frequent neighbouring pair: frequent strings become one token	GPT-2 to 4, LLaMA 3
WordPiece	glue the pair that most raises the training text's likelihood: the vocabulary predicts the corpus	BERT
SentencePiece unigram	start with many candidate pieces; prune those whose removal costs least likelihood	T5, XLM-R

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- All three learn from counts on their own corpus, not from meaning: one phrase, three cuts (Sennrich et al. 2016; Schuster & Nakajima 2012; Kudo & Richardson 2018).
- **The pieces are a property of the model that learned them, not of the language.**

One Sentence, Three Tokenizers



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- cl100k falls back on raw computer code for a character it saw rarely (red): 央 comes out as two meaningless pieces, but the sentence still has a representation.
- English BERT knows few Chinese characters: 9 of its 12 tokens are [UNK], “unknown”, one token for every character not on its list: the sentence is lost. Chinese BERT cuts every character into its own token.
- Right column: the same two sentences in English, 10 tokens to cl100k, 20 to Chinese BERT, which spells English from fragments.
- **The same sentence is a different number of tokens to each model.**

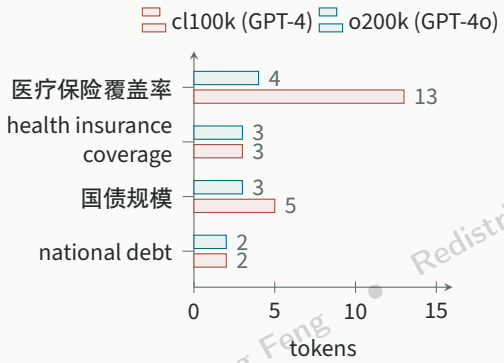
The Tokenizer Belongs to the Model



- One term, four cuts: 3 pieces, 8, 6 characters, or [UNK]s. “GDP rose by 2%”, 5 tokens cut by words, is 6 or 7 GPT tokens.
- A model’s first layer is an **embedding table**: one learned vector per token ID. Feed it another tokenizer’s IDs and every row is the wrong vector.
- Mixed Chinese–English notes and filings that mix 简 and 繁 call for byte-level BPE.
- **Report the tokenizer, and its version, as you report the sample period.**

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Translate, or Read in the Original?



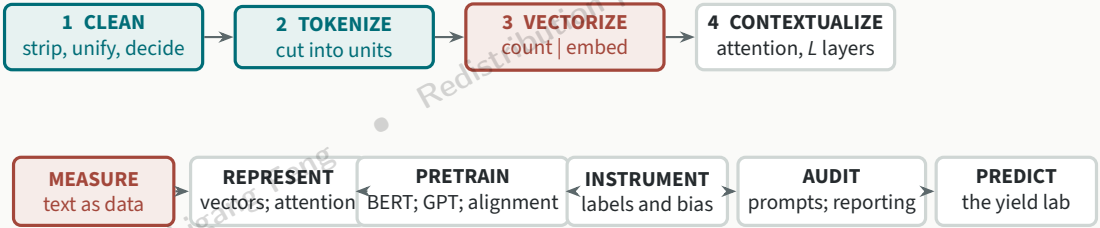
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- Translate a Chinese corpus and use English tools, or let a multilingual model read the Chinese?
- Translation loses shades: 稳健 is both stable and sound, “prudent” only caution; 适度宽松 is a named stance, not an adjective.
- Reading the original costs tokens; each bar is one phrase cut by one tokenizer: 医疗保险覆盖率 is 13 tokens to GPT-4’s, the English 3; GPT-4o’s cuts it to 4.
- **If a phrase is hard to translate, run both routes and report both.**

Session 5.1

Step 3 · Count

Turn tokens into vectors: one axis per word



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From Tokens to Numbers: IDs and One-Hot Vectors

vocabulary, numbered

	fed	cut	hike	support	curb	growth	inflation	respond
ID	1	2	3	4	5	6	7	8
$\boldsymbol{o}_{\text{fed}}$	1	0	0	0	0	0	0	0
$\boldsymbol{o}_{\text{hike}}$	0	0	1	0	0	0	0	0

$\boldsymbol{o}_{\text{fed}}^\top \boldsymbol{o}_{\text{hike}} = 0$: no two different words share a coordinate

- Number the vocabulary, $v = 1, \ldots, V$. Token v becomes the **one-hot vector** $\boldsymbol{o}_v \in \{0,1\}^V$: a 1 in place v, 0 everywhere else.
- For two different words $\boldsymbol{o}_v^\top \boldsymbol{o}_{v'} = 0$: every pair is **orthogonal** by construction. *hike* and *raise* are as far apart as *hike* and *cut*.
- **A document is the sum of its one-hot vectors: a row of counts.**

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Bag of Words and the Document–Term Matrix



three documents, $V = 8$: $\mathbf{Z} \in \mathbb{R}^{3\times 8}$, frequency counts

- Three short documents, already cleaned and tokenized: d_1 = fed cut support growth; d_2 = fed hike curb inflation; d_3 = fed hike respond inflation inflation.
- Count each word and ignore order: the **document–term matrix** $\mathbf{Z} \in \mathbb{R}^{D\times V}$, one row per document. Binary counts record presence (inflation in d_3: 1); frequency counts, how often (2).
- **Each row is its document's vector $\mathbf{z}_i$.**

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TF-IDF, Worked Through

term	df_v	$\mathrm{idf}_v = \log(D/\mathrm{df}_v)$	$\mathrm{tf}_{v,3}$	**TF-IDF in document 3**
fed	3	$\log(3/3) = 0$	1	0
hike	2	$\log(3/2) = 0.405$	1	0.405
inflation	2	$\log(3/2) = 0.405$	2	0.811
respond	1	$\log(3/1) = 1.099$	1	1.099

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- $\text{TF-IDF}_{v,i} = \mathrm{tf}_{v,i} \times \log(D/\mathrm{df}_v)$: the count of word v in document i, times the natural log of how rare the word is across documents. d_3 becomes $(0, 0, 0.405, 0, 0, 0, 0.811, 1.099)$.
- A graded stop list: **template text**, wording every document repeats, such as "financial statements", has $\mathrm{df} \approx D$ and a weight near 0, with no list written in advance.
- **The trap: if every statement contains "not", its weight goes to zero too.**

The Bag Forgets Word Order

GDP growth exceeded inflation

inflation exceeded GDP growth

both become the same count vector:



the rate cut supports growth

growth supports the rate cut

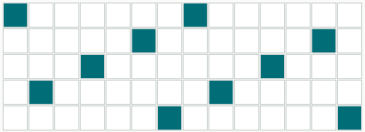
the same bag

- "GDP growth exceeded inflation" and its reverse get one vector; "the rate cut supports growth" and "growth supports the rate cut" state opposite causal directions in identical bags.
- Harmless when you count topic words, as the uncertainty index does; fatal when direction, cause or negation is the object.
- **Word order is gone. Deck 5.2 puts it back with position vectors.**

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Sparse: Most of the Matrix Is Zero

a document–term matrix, schematic



filled cells are non-zero; in the lab's 245 $\times$ 1,163 TF-IDF matrix, 85.5% of the cells are 0

- One corpus, three vocabularies so far, each a choice: 1,628 distinct tokens from the baseline pipeline; 1,163 TF-IDF columns (in two or more window statements, no numerals); 719 for the topic model (no stop, short or rare words).
- With fewer documents than words ($D \ll V$), OLS on all V columns is not identified: $\mathbf{Z}^\top\mathbf{Z}$ cannot be inverted. Reduce the dimension (topic models) or penalise (the Lasso).
- **That is why text regressions borrow the tools of machine learning.**

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Comparing Documents: Cosine on Counts

	fed	cut	hike	support	curb	growth	inflation	respond	raise
d_1	1	1	0	1	0	1	0	0	0
d_2	1	0	1	0	1	0	1	0	0
d_3	1	0	1	0	0	0	2	1	0
d_2'	1	0	0	0	1	0	1	0	1

$d_2 \cdot d_3 = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 2 = 4$
$\|d_2\| = \sqrt{1+1+1+1} = 2$, $\|d_3\| = \sqrt{1+1+2^2+1} = \sqrt{7}$
$\cos(d_2, d_3) = 4/(2\sqrt{7}) = 0.756$; $\cos(d_1, d_2) = 1/(2 \cdot 2) = 0.25$
$\cos(d_2', d_3) = (1+2)/(2\sqrt{7}) = 0.567$, d_2' is d_2 with *raise* for *hike*

- Words are orthogonal; documents are not. A row is a sum of one-hots, so in the **cosine similarity** $\mathbf{z}_i^\top \mathbf{z}_j / (\|\mathbf{z}_i\| \|\mathbf{z}_j\|)$ every word two documents share adds to the numerator.
- d_1, d_2 share only *fed*: $1/(2 \cdot 2) = 0.25$; d_2, d_3 share three words: 0.756.
- Swap *hike* for *raise* in d_2: same meaning, but the cosine with d_3 falls from 0.756 to 0.567.
- **Counts see shared strings, not shared meaning; Deck 5.2 repairs this.**

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Three Weightings of the Same Two Documents

d_2 = fed hike curb inflation
d_3 = fed hike respond inflation inflation

	fed	cut	hike	support	curb	growth	inflation	respond
counts d_2	1	0	1	0	1	0	1	0
d_3	1	0	1	0	0	0	2	1
presence d_2	1	0	1	0	1	0	1	0
d_3	1	0	1	0	0	0	1	1
TF-IDF d_2	0	0	.41	0	1.10	0	.41	0
d_3	0	0	.41	0	0	0	.81	1.10

$\cos(d_2, d_3)$: counts $4/(2\sqrt{7}) = 0.756$; presence $3/(2 \cdot 2) = 0.75$; TF-IDF $0.49/(1.24 \cdot 1.42) = 0.28$

- **Counts**: $z_{iv} = n_{iv}$. **Presence**: $z_{iv} = \mathbf{1}[n_{iv} > 0]$. **TF-IDF**: $z_{iv} = n_{iv}\,\mathrm{idf}_v$, $\mathrm{idf}_v = \log(D/\mathrm{df}_v)$ from the fitting documents.
- Three questions for a cosine: same words in the same amounts? same words at all? same *unusual* words?
- Template text (here *fed*) counts in full, once, or not at all: d_1, d_2, sharing only *fed*, score 0.25, 0.25, 0.
- **Same documents, three cosines: every cosine inherits the choice.**

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The Data: 245 FOMC Statements, One Template Edited Each Meeting

29 Oct. 2008

… decided today to lower its target for the federal funds rate 50 basis points to 1 percent.

In light of the declines in the prices of energy and other commodities and the weaker prospects for economic activity, the Committee expects inflation to moderate in coming quarters to levels consistent with price stability.

16 Dec. 2008

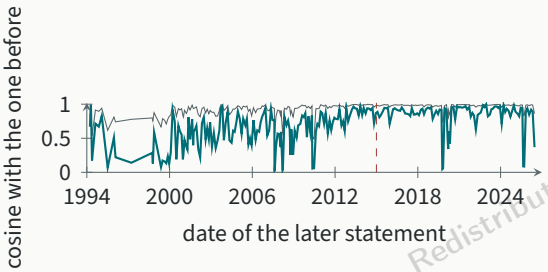
… decided today to establish a target range for the federal funds rate of 0 to 1/4 percent.

In light of the declines in the prices of energy and other commodities and the weaker prospects for economic activity, the Committee expects inflation to moderate further in coming quarters.

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- The corpus: every FOMC post-meeting statement, 1994-02-04 to 2026-06-17 (245 texts). Each is drafted from the one before: grey carried over word for word, red changed.
- Next: how much did the message change meeting to meeting? The cosine with the statement before, under each weighting; 10 Aug. 2007, a liquidity notice with no decision, is the odd text.
- **Carried-over template text: counts weigh it in full, TF-IDF near zero.**

Cosine on Real Statements: Three Weightings, Three Answers



pair	counts	0/1	TF-IDF
mean, 244 pairs	0.93	0.80	0.70
share above 0.9	79%	39%	28%
Oct. → Dec. 2008	0.81	0.55	0.26
7 → 10 Aug. 2007	0.56	0.26	0.02

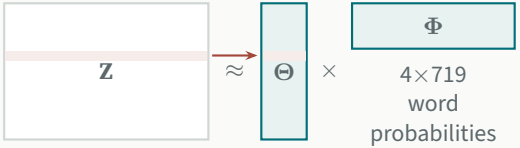
dashed line: vocabulary and IDF fitted on the statements left of it

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- x: the later date; y: cosine with the statement before; grey counts, teal TF-IDF.
- Counts: a typical pair scores 0.93, 79% lie above 0.9, Dec. 2008's rewrite 0.81: the template dominates. TF-IDF discounts it: mean 0.70, 28% above 0.9, and its dips are the meetings whose news changed (Dec. 2008 0.26; the Aug. 2007 notice 0.02).
- **Choose the weighting by the question: "same wording?" is counts, "same news?" is TF-IDF.**

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Topic Models: What We Have and What We Want

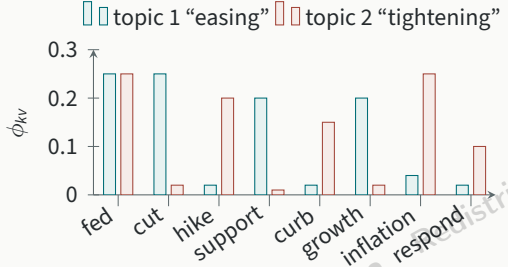


the red row: one statement, 719 counts on the left, 4 topic shares on the right

- What we have: **Z**, one row of 719 counts per statement: too many to regress on. LDA compresses them, as a factor model does.
- What we want: $K = 4$ topics, each a list of word probabilities ϕ_{kv} (what the topic says), and per statement K shares θ_{ik} (how much of each it uses).
- How: assume a story (next frame); guess each word's topic, improve the guesses, count.
- **Text enters a regression as K shares, not V counts.**

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The Assumption: How a Document Is Written, Word by Word



d_3 with $\boldsymbol{\theta}_3 = (0.1,\ 0.9)$, one draw:

hike	inflation	respond	inflation	fed
$z = 2$	$z = 2$	$z = 2$	$z = 2$	$z = 1$

- **Latent Dirichlet allocation** (Blei, Ng & Jordan 2003) assumes a story. Top: topic k's **distribution over words**, its bars ϕ_k.
- Two kinds of dice: each document its own K-sided die, its **topic shares** θ_i (d_3: 0.1, 0.9); each topic one V-sided die, its bars ϕ_k, shared by all. Per slot: roll the document's die for a topic z, then die z for a word w.
- **A document mixes topics; word order plays no part.**

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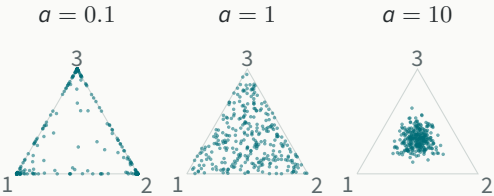
LDA's Symbols: Two Unknowns, Two Count Tables

symbol	what it is	size	toy	FOMC fit
z	the **tag** on one token: the topic that produced it	1 per token	*fed* in d_3: $z = 1$	21,673 tags
ϕ_k	the **bars**: topic k's distribution over words (the chance it produces each word); sums to 1	V per topic	the two bar charts	4×719
θ_i	the **shares**: document i's distribution over topics (the fraction of its words from each); sums to 1	K per document	$\theta_3 = (0.1,\ 0.9)$	149×4
$n_k^{(v)}$	the **word table**: tokens of word v now tagged k; a row, normalised, is $\hat{\phi}_k$	$K \times V$ counts	worked frame, top	4×719
$n_i^{(k)}$	the **document table**: tokens of document i now tagged k; a row, normalised, is $\hat{\theta}_i$	$D \times K$ counts	worked frame, bottom	149×4
$a,\ b$	pseudo-counts added before normalising, so no share or bar is ever 0	two numbers	$0.1,\ 0.01$	$0.1,\ 0.01$

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- Fitting runs the story backward: words $\rightarrow$ every tag $z \rightarrow \hat{\phi}, \hat{\theta}$ read off the two tables.
- **Bars and shares are the unknowns; the two tables are the counts behind them.**

Why “Dirichlet”: A Knob on How Focused a Document Is



$K = 3$; a corner = all words from that topic; a point = one document’s shares θ; 300 draws each. $a = 0.1$: 67% of draws give one topic more than 0.9; $a = 10$: in 94% every share is within 0.2 of $1/3$.

- The **Dirichlet** is a distribution over shares: each draw is a whole die. LDA draws each θ_i from $\mathrm{Dir}(a)$, each ϕ_k from $\mathrm{Dir}(b)$.
- Small a: one or two topics per document; large a: all of them (left). b: the same for a topic’s words.
- The same a as in the read-off, $\hat{\theta}_{ik} = (n_i^{(k)} + a)/(n_i + Ka)$: a pretend count added to every topic.
- **FOMC fit: $a = 0.1$, $b = 0.01$: few topics per statement, few words per topic.**

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Fitting LDA: The Question and the Idea

a current guess: every token carries a topic tag

d_1 fed cut support growth

d_2 fed hike curb inflation

d_3 fed hike respond inflation inflation

teal: topic 1; red: topic 2; white: the token re-guessed. Were these tags right, fitting is counting (next frame): 8 red tokens, 3 of them *inflation*: $\hat{\phi}_{2,\text{inflation}} = 3/8$; d_3's other tags all red: $\hat{\theta}_3 \approx (0, 1)$.

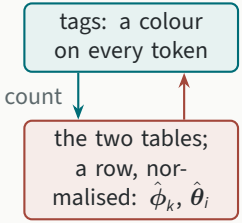
which topic likes *fed*?

which topic does d_3 use?

- We see only words and want each token's tag z; with the tags known, the bars $\hat{\phi}$ and shares $\hat{\theta}$ are the frequencies on the left.
- Circular: tags need ϕ, θ, which need tags. The way out: hold every other tag fixed and re-guess one, from the two questions below left.
- Token by token, round after round, the guesses settle on the posterior (**Gibbs sampling**).
- **"Collapsed": ϕ, θ are never drawn, only counted.**

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The Loop: Tags Give Counts, Counts Judge Tags



down, as drawn on the previous frame: 8 red tokens, 3 of them *inflation*: $\hat{\phi}_{2,\text{inflation}} = 3/8$; d_3 all red: $\hat{\theta}_3 = (0, 1)$. Up, *fed* in d_3 drawn teal (1 in 20): $n_1^{(\text{fed})}$ $1 \to 2$, $n_1^{(\cdot)}$ $4 \to 5$, $n_3^{(1)}$ $0 \to 1$, so $\hat{\phi}_{1,\text{fed}} = 1/4 \to 2/5$, $\hat{\theta}_3 = (0, 5)/5 \to (1, 4)/5$; drawn red: nothing moves (*a*, *b* left out here).

- Down: with the tags in place, bars and shares are the tables' rows, normalised; nothing else is estimated.
- Up: no answer key exists. A tag is judged only by the other tags, through the tables: does topic *k*'s row favour this word, does this document's row favour *k*?
- A re-drawn tag moves one count in each table, and bars and shares move with it (left); they are never set by hand.
- **Stop when the tags agree with each other; the thermometer says when.**

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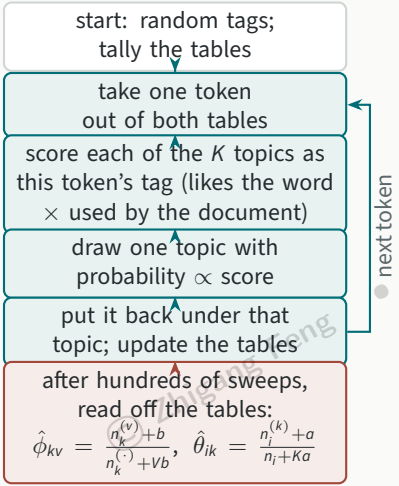
Gibbs Sampling in Words: Labels That Correct Each Other



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- Gibbs sampling turns words into tags; $\hat{\phi}$, $\hat{\theta}$ are just the tags counted (below each panel).
- One token at a time: hide its colour; ask which colour this word carries elsewhere and which this document's other words carry; multiply, roll a weighted die, colour it.
- Why it settles: words that keep appearing together lift each other's chance of one colour, in and across documents; a lone mistake has no support and is overwritten.
- The log-likelihood is a thermometer, not the engine: when it stops rising, stop and count.
- **No one labels by hand: each label is re-guessed from all the others until they agree.**

Collapsed Gibbs Sampling: Start, One Step, Stop



1. **Start.** Random tags; tally the two count tables.
2. **One step**, for word v in document i, tagged k: subtract 1 from $n_k^{(v)}$, $n_k^{(\cdot)}$, $n_i^{(k)}$ (its own tag must not be evidence); for every topic, s_k = how plausible it is that topic k produced this token:

$$s_k = \underbrace{\frac{n_k^{(v)} + b}{n_k^{(\cdot)} + Vb}}_{\hat{\phi}_{kv}\text{: topic } k\text{'s share of } v} \underbrace{\left(n_i^{(k)} + a\right)}_{\propto\, \hat{\theta}_{ik}\text{: } i\text{'s tokens tagged } k}$$

   $p_k = s_k / \sum_j s_j$; draw $u \sim U[0,1]$, the first k with $p_1 + \cdots + p_k > u$ wins; add 1 to the same three counts for that k.
3. **Stop.** A **sweep** visits every token once; run hundreds; read off (red box).

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One Step, Worked: the Toy's Two Tables, Then *fed* in d_3

word table, all three documents pooled: row = topic, column = word, cell = count

	fed	cut	hike	support	curb	growth	inflation	respond	
$k=1$	1	1	0	1	0	1	0	0	$n_1^{(\cdot)}=4$
$k=2$	2	0	2	0	1	0	3	1	$n_2^{(\cdot)}=9$

document table: tokens tagged 1 (teal) and 2 (red), per document

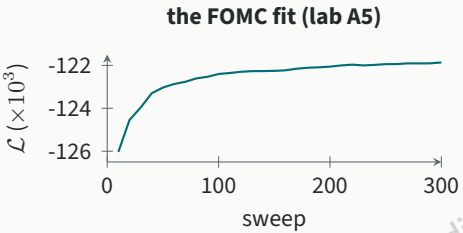
d_1	4	0	d_2	0	4	d_3	0	5

after the draw, the counts that change: red (19 in 20): back to $n_2^{(\text{fed})}=2, n_2^{(\cdot)}=9$, $n_3^{(2)}=5$; teal (1 in 20): $n_1^{(\text{fed})}=2, n_1^{(\cdot)}=5$, $n_3^{(1)}=1$, topic 2 keeps 1, 8, 4

- **Out.** *fed* in d_3 is red; subtract 1: $n_2^{(\text{fed})}$ $2\to1$, $n_2^{(\cdot)}$ $9\to8$, $n_3^{(2)}$ $5\to4$. Every other tag, *fed* in d_1 and d_2 included, stays and votes.
- **Scores** ($a=0.1$, $b=0.01$ our choice; $V=8$): $s_1=\frac{1.01}{4.08}\times0.1=0.025$ (likes *fed*, 1 of 4; d_3 never uses it); $s_2=\frac{1.01}{8.08}\times4.1=0.513$ (1 of 8; d_3 uses it four times). $p=(0.05,\ 0.95)$.
- **Draw, put back.** Red, 19 times in 20: add 1 back, the counts return to 2, 9, 5. Teal, once in 20: three topic-1 counts rise instead (left); the next token sees the new tables.
- **Draw; never take the largest p_k: that one chance in twenty is how the chain explores.**

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The Thermometer: How Probable Are the Words Under the Current Fit?



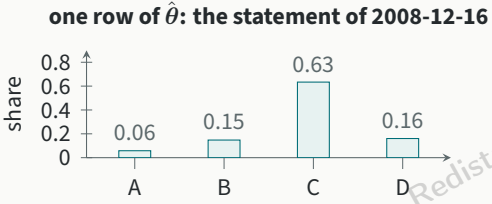
x = sweep; $y = \mathcal{L}$ computed from the tables after that sweep: it rises from -126,029 and is flat by 300 sweeps (-121,864).

$$\mathcal{L} = \sum_{\text{documents } i} \sum_{\text{tokens } v \text{ of } i} \log \underbrace{\sum_k \hat{\theta}_{ik}\, \hat{\phi}_{kv}}_{\text{chance of this word in this document}}$$

- Inside the log: $\sum_k \hat{\theta}_{ik}\hat{\phi}_{kv}$, the products one step scores with; p_k is topic k's share of the sum. Log; add over every token.
- Nonsense tags: every topic a blur of all words, every document of all topics; each observed word improbable: $\mathcal{L}$ low.
- Tags sorting themselves: topics concentrate on their words, documents on theirs; words more probable: $\mathcal{L}$ rises, then flat: stop.
- **We never climb it; we read it.**

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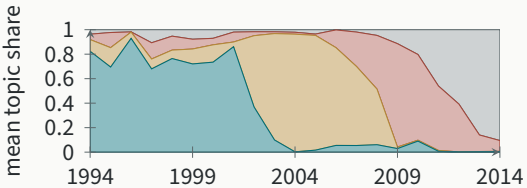
Lab A5: Topics in the FOMC Statements, the Setup



- Can 4 shares stand in for 719 counts in the 1994–2014 statements?
- The thermometer (previous frame) is flat by 300 sweeps: the chain has settled, so we read the tables.
- Left: $\hat{\theta}$ of the 2008-12-16 statement: x = topic, y = share of its tokens; 0.63 in C.
- **The fit:** $K = 4$**,** $a = 0.1$**,** $b = 0.01$**, 300 sweeps.**

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LDA on the FOMC Statements: What It Found

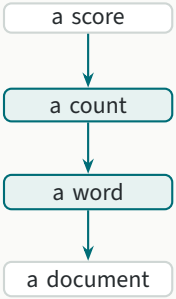


top words, and the name we gave
reserve, discount, board, available *A: discount-rate announcements*
chairman, risks, decided, voting *B: the vote and the balance of risks*
financial, markets, levels, housing *C: crisis, markets and housing*
longer, term, consistent, pace *D: longer-run goals and purchases*

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- Area: x = year, y = mean $\hat{\theta}$ over the year's statements, bands stacking to 1. Table: each topic's four largest $\hat{\phi}_{kv}$. Topics come back numbered, not named: the names are ours.
- Four drafting eras (A to 2002, B from the 2002 vote roster, C 2009–11, D 2012–14): the word clusters track how the statement was written more than what the economy did.
- **Read a topic's words and its time path before its share enters a regression.**

When Counting Still Wins

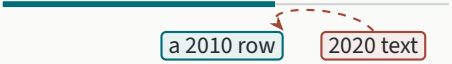


every coordinate is a word, so a score can be traced back to its source

- When the question is how often defined words occur, transparency is the value: the uncertainty index, finance sentiment lists, political risk.
- An audit traces a score back to its words and documents; a dense vector cannot.
- Cosines of binary word vectors of 10-K product descriptions define product-market rivals (Hoberg & Phillips 2016); the same can be tried on the business sections of other annual reports.
- **Under about a thousand documents, TF-IDF is the sound baseline.**

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No Look-Ahead: Fit the Weights on Past Statements Only

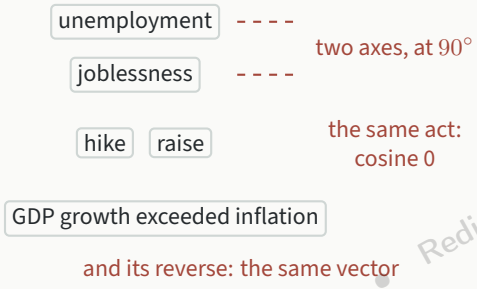


	fit on 1994–2014		fit on all 245	
word	df	IDF	df	IDF
balance	33	1.51	89	1.01
securities	49	1.11	120	0.71
pandemic	0	no column	19	2.56

- Which documents does df_v in $\log(D/\mathrm{df}_v)$ count? Left: two answers.
- Over all 245: *balance* is in 56 of the 96 later statements, so a 2010 row weighs it 1.01, not 1.51: a 2010 number built from 2020 texts.
- **Look-ahead**: a 2010 forecast built on it (Deck 5.6) uses texts no one had in 2010. The rule: fit on 1994–2014, apply unchanged; *pandemic* gets no column.
- **"Fitted on the future": a df that counts documents dated after the row.**

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What Counts Cannot See



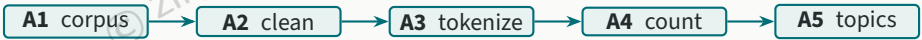
- Cleaning, tokenizing and counting turned each document into a row of counts: transparent, auditable, and blind to meaning.
- Synonyms sit on perpendicular axes, word order is gone, and every word is its own axis.
- Deck 5.2 gives up one axis per word: words become points, and nearness means similarity.
- **Counts tell you which words were used, not what was meant.**

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Lab, Part A: Clean, Tokenize and Count the FOMC Statements

cell	what you run	the question you answer
A1	load 245 statements; set the 1994–2014 window	How many statements may a fitted object see?
A2	five cleaning switches, one at a time	Which switch moves the vocabulary most, and what does it delete?
A3	byte-pair encoding, 200 merges	How is a word the window never saw, such as *reflationary*, cut?
A4	document–term matrix and TF-IDF, fitted on the window	How much held-back text falls outside the vocabulary? What would a full-sample IDF change?
A5	LDA with $K = 4$, collapsed Gibbs	Your names for the topics, a neighbour's, and what changes with seed 7 or $K = 6$?

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- **Every number on the counting slides is one you can reproduce.**